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Is exposure to chemical pollutants associated with sleep outcomes? A systematic review

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Summary:

Environmental exposures may influence sleep; however, the contributions of environmental chemical pollutants to sleep health have not been systematically investigated. We conducted a systematic review to identify, evaluate, summarize, and synthesize the existing evidence between chemical pollutants (air pollution, exposures related to the Gulf War and other conflicts, endocrine disruptors, metals, pesticides, solvents) and dimensions of sleep health (architecture, duration, quality, timing) and disorders (sleeping pill use, insomnia, sleep-disordered breathing)). Of the 204 included studies, results were mixed; however, the synthesized evidence suggested associations between particulate matter, exposures related to the Gulf War, dioxin and dioxin-like compounds, and pesticide exposure with worse sleep quality; exposures related to the Gulf War, aluminum, and mercury with insomnia and impaired sleep maintenance; and associations between tobacco smoke exposure with insomnia and sleep-disordered breathing, particularly in pediatric populations. Possible mechanisms relate to cholinergic signaling, neurotransmission, and inflammation. Chemical pollutants are likely key determinants of sleep health and disorders. Future studies should aim to evaluate environmental exposures on sleep across the lifespan, with a particular focus on developmental windows and biological mechanisms, as well as in historically marginalized or excluded populations.

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Keywords

Environmental Pollutants; Sleep Health; Sleep Wake Disorders; Sleep Initiation and Maintenance Disorders; Particulate Matter; Dioxins; Endocrine Disruptors; Pesticides; Heavy Metals; Mercury; Solvents; Cholinergic Agents; Gulf War Illness

INTRODUCTION

Sleep is a dynamic neurophysiological process regulated by a 1) homeostatic sleep drive and 2) a circadian rhythm in wakefulness[1]. Healthy sleep is a multidimensional construct defined by sleep duration, efficiency, timing, alertness, and quality[2]. However, disrupted or impaired sleep[3] and sleep-related disorders such as obstructive sleep apnea (OSA)[4] or chronic insomnia[5] are common. The pervasiveness of these outcomes is alarming because sleep is essential for metabolic, immunologic, developmental, and cognitive functioning[6].

In addition to structural and sociodemographic factors, health status, and individual behaviors, environmental factors are particularly important for sleep health[7]. Prior systematic reviews of environmental exposures and sleep have summarized associations with air pollution[8], second-hand smoke (SHS) exposure[9,10], and occupational exposures[11]. However, despite biological plausibility, less is known regarding the effects of chemical pollutants such as pesticides, heavy metals, solvents, and endocrine-disrupting chemicals (EDCs) on sleep health. Chemical pollutants are common environmental exposures that may disturb sleep by acting upon the biological pathways that regulate sleep-wake behavior, with developmental windows of vulnerability. However, the connections between environmental chemical exposures and sleep disturbance are not well-established. Importantly, environmental pollution does not affect all people equally; because of environmental racism, marginalized communities face a disproportionately higher burden of adverse environmental exposures, such as environmental disparities in air pollution[12], which may contribute to sleep health disparities[13,14].

Chemical pollutants may contribute to sleep disruption by influencing the underlying biology of sleep-wake behavior. Therefore, to address these gaps in knowledge and to synthesize the current epidemiological evidence, we conducted a systematic review of the relationships between chemical pollutants and sleep health and disorders. Furthermore, we discuss potential mechanisms linking pollutants to sleep outcomes and suggest areas for future research.

METHODS

Data Sources and Literature Search Strategy

We used the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines in conducting this review and registered our protocol with the international prospective register of systematic reviews, PROSPERO (#CRD42021256918, first submitted May 27, 2021). As outlined by our protocol, a comprehensive search was created by the research team and executed by information specialists (SP and KF) in 6 bibliographic databases: Agricultural & Environmental Science Collection (ProQuest),

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PubMed, Embase (Elsevier), Environment Complete (Ebsco), Web of Science Core Collection, and Scopus (Elsevier) in the Spring of 2021. Search terms, dates, and additional details are provided in Supplemental File 1. The database search retrieved 9,918 citations. Studies were imported into EndNote X9 for data management and deduplication. Additional duplicates were removed by the web-based evidence synthesis tool Covidence.

Inclusion criteria

Inclusion criteria included: research in humans, chemical pollutant as an exposure, sleep health or disorder, sleep measurement, and quantitative data. Studies of *in vitro* or cultured tissue, completely modeled data, or non-human data were excluded from the analysis; however, in support of the epidemiological literature and biological plausibility, we mention relevant findings from experimental animal research in the Discussion. Studies of non-chemical environmental pollutants, such as noise, were excluded from the analysis. A recent systematic review by Liu et al[8] summarized air pollution literature (n=22 articles) up until October 2019; therefore, we excluded these previously reviewed studies[8] and discuss their relation to our findings. We excluded studies of micronutrients and sleep except a few with relevance to pesticide use and contamination: manganese, copper, and selenium. Symptoms of sleep disorders such as snoring, restless legs syndrome, and rapid eye movement (REM) sleep behavior disorder, as well as diseases such as Parkinson's disease, were excluded due to being outside the review's scope. Studies with only qualitative data, not available in English, not peer-reviewed, or epidemiologic studies without a control or comparator group were also excluded. Inclusion categories were further reviewed during mutual group discussions (DW, JPG, DAJ).

Screening

The resulting 4,464 unique identified papers were screened by two independent reviewers (DW and JP) against inclusion and exclusion criteria using Covidence (Figure 1). After the full-text screening, bibliographies of 20 review articles (Supplemental File 2) were searched for additional citations in February 2022, resulting in an additional 58 articles (2 duplicates) added to Covidence for title/abstract and full-text screening. Of the initial papers, 3,984 articles were considered irrelevant and excluded during the title/abstract screening process. Full text review of the remaining 538 papers resulted in 204 studies included for data extraction, 15 of which were identified from the review articles. Of the included articles, 74% (n=151) were epidemiological studies and 26% (n=53) were case reports or case series. Consensus was arbitrated by the first author during title/abstract and full-text screening, and, in the instance of discordant responses, consensus was determined by mutual discussion (DW and JPG).

Data Extraction

Data on year, study location, study design, exposure and exposure measurement, outcome and outcome measurement, and risk of bias (RoB, details below) for the 204 included studies were independently extracted by two reviewers (DW and JPG) using Covidence and Google Forms (more details and abstraction forms in Supplemental File 3). Results were merged, and when discordant, consensus determined by mutual discussion. Only results pertaining to ambient or SHS exposure, and not active smoking or *in utero* exposure, were

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included. Due to heterogeneity, meta-analysis and comparison of effect estimates was not performed.

Risk of Bias (RoB) Assessment

Included epidemiologic studies were rated for risk of bias (RoB) according to questions adapted from both the National Institute of Environmental Health Sciences and National Toxicology Program's Office of Health Assessment and Translation tool (https://ntp.niehs.nih.gov/ntp/ohat/pubs/riskofbiastool_508.pdf) and the Cochrane ROBINS-I tool (template tool available in Supplemental File 4). RoB was independently assessed by two reviewers (DW and JPG) using Covidence and Google Forms; when responses differed, consensus was reached after discussion. Overall RoB was assessed first by intervention domains and then weighted across the selection bias, confounding, exposure assessment, and outcome assessment domains. For case reports and case series, RoB was graded according to a tool proposed by Murad et al[15] (more details and template tool available in Supplemental File 4). Results were visualized using R version 4.1. Data extraction and quality assessment responses are provided in Supplemental File 5 and Supplemental File 6.

RESULTS

The included epidemiologic studies (n=151) evaluated exposure to air pollution (19%), conflict-related exposures (10.5%), EDCs/other (8%), metals (34%), pesticides (8%), solvents (20.5%), and exposures with sleep outcomes related to sleep architecture (2.6%), dreams/nightmares (6%), duration (26.5%), quality (47.7%), timing (2.6%), sleeping pill use (4%), sleep maintenance/insomnia (37.1%), and SDB (23.8%), as shown in Supplemental File 7A. Most case studies (n=53) reported exposure to metals (64%) and sleep outcomes such as insomnia (50.8%) and quality (29.2%)(Supplemental File 7B). Only 17.6% of epidemiologic and case studies including pediatric populations, and of the 22 epidemiologic and 14 case study articles with pediatric populations included in this review, the most common exposures were metals (47%) and/or air pollution (28%), particularly SHS. Most epidemiologic studies relied on device-based measures of the exposure (61%) and self-reported measures of sleep (77%)(Supplemental File 7C) using questionnaires such as the Pittsburgh Sleep Quality Index (PSQI, N=16). PSG was the most common device-based measure (N=17), and only 5 studies assessed sleep via actigraphy with commercially-available wrist-worn devices; 2 additional studies relied upon static charge sensitive beds.

Air Pollution

Air pollutants comprised: particulate matter (PM_1, $PM_{2.5}$, PM_{10}), nitrogen oxides (NO_x), ozone (O_3), sulfur oxides (SO_x), secondhand smoke (SHS) or environmental tobacco smoke (ETS), and volcanic ash (Supplemental Files 8 and 9). Epidemiologic studies of air pollution (n=29) mostly occurred in urban environments in Asia and Europe, relied on device-based measures of exposure (65.5%), and evaluated self-reported measures (79%) of sleep quality or insomnia. Objective measures of ambient air pollution exposure were non-specific, captured with outdoor monitoring stations and land use regression models; one study utilized 24-hour indoor air measurement[16]. No studies employed wearable

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sensors or personal air monitoring devices in objective exposure assessment; person-specific exposures were ascertained from self-report and/or from metabolite measurement.

There was general support for associations between ambient air pollution and disrupted sleep. Findings from the Henan Rural Cohort Study reported associations between PM and nitrogen dioxide (NO_2) with longer sleep latency[17] and worse sleep quality[18,19]; in these studies, air pollutant exposure was averaged across the 3 years prior to the PSQI questionnaire to evaluate the association of chronic air pollution exposure to self-reported sleep quality collected at a single time-point. PM and NO_2 levels, averaged over one year, were also linked to incident sleep disorders in a prospective cohort study in China[20]. However, smaller studies in Europe and the U.S. did not replicate these findings[21,22]. One with 6 weeks of actigraphy measurement evaluated within-person associations between daily O_3, $PM_{2.5}$, SO_2, NO_2, CO and wake after sleep onset, sleep efficiency, sleep duration, and self-reported sleep quality, and reported an association between daily maximum ozone levels and longer sleep duration[21]. A study of UK Biobank data with air pollution data measured across a few years reported an association between NO_x and 7% increased odds of sleep disorders (which included insomnia and sleep apnea) when modeled as a singular pollutant, but a protective effect between NO_2 and sleep disorders when $PM_{2.5}$, PM_{10}, and NO_x were included as covariates in the model[23]. The few pediatric studies of ambient air pollution investigated SDB, with heterogenous findings[16,24,25], such as an ecological study in Italy reporting higher NO_2 exposure and SDB in children[24]. While this study used a device-based measure of SDB, it is limited by an ecological study design and use of AHI >1 event per hour (often used as a pediatric cutoff for mild SDB) as the threshold to define the SDB outcome [24]. Air Quality Index (AQI) was also found to be associated with insomnia in an ecological study[26] and admission for sleep disorders[27]. The overall quality of the included studies was mixed (Supplemental File 10A).

Of the 5 studies of indoor or household air pollution (HAP), the study with device-based air measurement reported null findings with SDB in children[16]; 2 studies reported associations between questionnaire-assessed solid fuel use and poor sleep quality[28] and biomass use and SDB and sleep disruption[29]; in these studies, chronic exposure to solid fuel was ascertained by asking participants to report use over the last 5 years or fill in the number of years used and weekly frequency, respectively. However, a larger study reported null findings between heating stove use and pediatric SDB[25]. One study with measures of urinary polycyclic aromatic hydrocarbon metabolites found an association between PAH exposure and poor sleep quality[30].

The majority of the 11 reviewed studies with ETS or SHS exposure reported positive associations between exposure and sleep disruption, although results differed by study population. Most assessed tobacco exposure through self-report, but of the 5 studies with measures of cotinine (a metabolite of nicotine), 2 reported positive associations with sleep impairment and other results were null. One of these studies measured SHS with both urinary cotinine and questionnaires and sleep via PSG[31] and found no association. ETS/SHS exposure measurement was heterogeneous; for example, some studies did not ask specifically about tobacco smoke exposure, but rather measured proximity to smokers or avoidance behavior[32]. In healthy children, self-reported SHS exposure was associated

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with increased Sleep-Related Breathing Disorder (SRBD) scores in school-aged children in Chile[33] and Turkey[25]. Additionally, in a cohort of healthy children ages 2–5 years with SHS measured by hair nicotine, there was an association with self-reported sleep-related breathing problems[34]. In children with asthma from Cincinnati ages 5–13 years, serum cotinine levels were associated with elevated SRBD scores and self-reported sleep problems including longer sleep onset delay, parasomnias, daytime sleepiness, and overall sleep disturbance[35]. However, in children ages 7–11 years with overweight or obesity, there was no association between SRBD scores and passive smoke exposure when measured by plasma cotinine assay[36]. Overall, there's good evidence that SHS exposure is associated with poor sleep outcomes in healthy children and those with asthma; however, due to the cross-sectional design, a causal association for adverse sleep outcomes could not be ascertained. Most pediatric studies demonstrated an association between ETS or SHS and self-reported SDB, insomnia, and poorer quality sleep.

Conflict-related Exposures

There were 16 included epidemiologic studies with exposures related to the Gulf War and Gulf War Illness (GWI), the World Trade Center (WTC) toxic dust exposures of first responders, and other conflict-related exposures (Supplemental Files 8 and 11). All investigations of GWI relied on self-reported measures, whereas WTC studies relied on self-reported exposures or known occupational history with PSG or home sleep apnea test (HSAT) measures of SDB. Some of the included studies of GWI focused on the categorization and case definition of GWI, which included sleep symptoms[37–39]. In addition to insomnia and worse sleep quality, one study also reported a higher OSA STOP questionnaire score among Gulf War veterans with GWI, compared to those without GWI[40]. Of the WTC studies, 5 out of 7 did not support an association between exposure duration or dose and SDB[41–45]; however, the largest study of responders (n=11,701 total) reported an association between earlier arrival time to the WTC site and higher risk of incident OSA[46]. Follow-up studies reported similar findings, with earlier response time and exposure to the dust cloud associated with greater log apnea-hypopnea index (AHI)[44] and severe, but not mild or moderate, OSA[47]. Among the two studies of sarin exposure, an OP compound, a study by Duffy et al from 1979[48] reported differences in sleep architecture, such as increased REM sleep, among male workers occupationally exposed to sarin. Among civilians, sarin exposure was also linked to a higher frequency of insomnia, but not bad dreams, 3 years following a sarin attack, compared to unexposed[49]. The only study of sulfur mustard exposure thoroughly evaluated sleep health using both validated questionnaires (PSQI and STOP-Bang) and device-measured sleep (PSG), as well as serum melatonin, reporting detrimental effects of occupational sulfur mustard exposure on all of these outcomes[50]. Study quality was mixed (Supplemental File 10B), with overall supporting evidence between Gulf War exposures, sarin, and sulfur mustard with impaired sleep quality, insomnia, SDB, and altered sleep architecture.

Endocrine Disrupting Chemicals (EDCs) / Other

This category included 12 epidemiologic studies and 3 case studies of EDCs and other chemicals that weren't well captured by other categories, such as bisphenol A (BPA), dioxins, per- and polyfluoroalkyl substances (PFAS), phthalates, polybrominated diphenyl

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ethers (PBDEs), and polychlorinated biphenyls (PCBs). Most sources of exposure to EDCs were unknown or occupational and had questionnaire-measured outcomes (Supplemental Files 8 and 12). BPA, as measured with mass spectrometry of urine samples collected at a single point in time, was protective of short sleep in an NHANES analysis, but this relationship did not remain across BPA quartiles[51]; because of the high within-day and day-to-day variability in urinary BPA[52], these results may be limited by the use of an acute BPA measurement for a chronic sleep outcome. In another study, patients with OSA had higher BPA levels (measured in serum at a single time point) compared to adults without OSA[53]. In a prospective birth cohort, female, but not male, children with the highest tertile of cord-blood measured PBDE had greater problems sleeping compared to those in the lowest tertile at four years old[54]. PCBs and dioxins were also linked to problems sleeping[55,56], although in one case, exposure was due to a fire and stress may be a confounder[56]. A different study of 140 cases of PCB exposure from contaminated rice bran (blood measured every few years) and followed over time reported worse sleep quality and higher prevalence of difficulty initiating sleep among those in the highest quartile of exposure[57]. Of the two studies of phthalates (measured in urine), an analysis of adults in the Midlife Women's Health Study did not report an association with insomnia or sleep disturbance except in former smokers; an analysis of NHANES participants ages 16–17 did report higher odds for short (<8 hours) weekday sleep duration among those in the highest quartile of phthalate exposure, but did not consider school start times or the influence of puberty and chronotype on the outcome[58]. The overall quality of studies varied (Supplemental Files 10C and 13), with no clear consistent associations, possibly compounded by the heterogeneity of the pollutants. However, persistent EDCs such as dioxins, PCBs, and PBDEs showed the most consistent evidence for sleep disruption.

Metals

Most epidemiologic (n=51) and case (n=34) studies of metals focused on exposure to lead or mercury, but a few also investigated aluminum, antimony, arsenic, cadmium, coal ash, copper, fluoride, lead, manganese, mercury, selenium, and thallium (Supplemental Files 8 and 14). Common exposure sources were workplace or occupational, particularly welding and/or metallurgy work, as well as dental work and amalgams. In the occupational settings, chronic exposure to metals was assessed with chronic sleep health outcomes such as insomnia. Among metals represented by more than one epidemiologic study, working with aluminum (measured in air, urine and/or occupational history) was linked to insomnia[59,60]. Studies of mercury and insomnia or sleep quality were inconsistent; however, positive associations were more often reported with higher occupational exposure to mercury[61,62], whereas most studies of dental amalgams and dentistry workers reported null associations[63–66]. In most of these studies of dental amalgams, exposure was assessed with occupational history or a questionnaire; however, a few included objective measures, such as measures of mercury vapor in the mouth during dental work[67] and air and urinary mercury[66]. In both adult and pediatric populations, exposure to lead was most consistently associated with insomnia[68–70] and shorter sleep duration[71–73], whereas associations with sleep quality were mixed. Thallium exposure was also linked to sleep disruption[74–76], although some of these findings may be confounded by smoking. The evidence between metals exposure and SDB were inconsistent. Associations between arsenic

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and sleep were null[77,78], and the majority of studies did not support an association between metals exposure and sleep duration in adult populations (Supplemental File 10D).

Pediatric findings were inconsistent. Most studies reported null associations with sleep quality and mixed associations with sleep timing, with positive sleep timing findings in relation to self-reported sleep[73,79] and null findings with actigraphy measured sleep[80,81]. A cross-sectional study reported that blood lead levels ≥10 μg/dL were associated with decreased sleep duration in Mexican children aged 6–8 years[73]. Longitudinal studies in China[70] and Mexico City[72] similarly reported that elevated blood lead levels in early childhood were associated with an increased risk of insomnia and excessive daytime sleepiness[70] and decreased sleep duration[72] in later childhood. A cross-sectional study of children ages 9 to 11 reported that higher blood mercury levels were associated with shorter sleep duration[80]. However, a recent cross-sectional study of Mexican children did not support an association between urinary mercury and sleep duration[81], although toxicokinetics and measurement in different biomatrices may explain discordant findings. Overall, there is evidence for negative consequences of early lead exposure and poorer sleep outcomes.

In contrast to the epidemiologic literature, most case studies in adults and children reported an association with insomnia and sleep quality and were of fair or good quality (Supplemental Files 13 and 14). A case study with multiple biological measures of mercury exposure and environmental sampling reported sleep disturbances among 70% of 18 adults living in a town polluted by mercury due to gold mining[82]. Other case studies reported symptoms of sleep disturbance among male welders following exposure to lead[83,84], manganese[85,86], and other metals from welding fumes. Overall, findings supported a link between exposure to mercury or lead and insomnia.

Pesticides

Epidemiologic (n=12) and case (n=5) studies of pesticide exposure had small sample sizes and included organophosphates (OPs), organochlorines, and/or carbamates (Supplemental Files 8 and 15) from a rural/farmland source in the Middle East, Africa, or North America. Some studies relied on farm work as a proxy for pesticide use and exposure; others did not explicitly state the pesticide class or chemical(s) evaluated. All of the studies relied on self-reported pesticide use and/or blood measures of cholinesterase and all but one[87] utilized self-reported measures of sleep. In one study, male pesticide applicators in the Agricultural Lung Health Study with self-reported exposure to carbamates, but not OPs, had higher self-reported doctor-diagnosed sleep apnea, with or without adjustment for BMI[88]. In a cross-sectional study of greenhouse farmers (n=1,336), farmers with medium or high cumulative exposure to pesticides had worse sleep quality and more trouble falling asleep compared to those with low exposure; highly exposed farmers also had 56% higher odds of short (≤6 hours) sleep duration[89]; however, results were not differentiated by pesticide class. An investigation of households in Ohio that had been illegally treated with OPs reported 28% of participants in the contaminated households experienced night waking, and in one case a 4-month-old infant with disrupted sleep was brought to the emergency room after their sleeping area was sprayed[90]. In another

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case, an adult male developed sleep apnea after improper treatment of his home with Carbaryl, a carbamate compound[91]. None of the included studies specifically evaluated a pediatric population. Most epidemiologic studies of pesticide exposure and insomnia reported an association[92–95], whereas associations with sleep duration, quality, and SDB were inconclusive. In summary, the overall epidemiologic evidence between pesticide exposure and sleep disruption was of low quality (Supplemental File 10E), in contrast with the overall evidence in case studies, which was good (Supplemental Files 13 and 15).

Solvents

Epidemiologic (n=31) and case (n=11) studies of solvents evaluated occupational exposure to benzene, toluene, and xylenes, among others (Supplemental Files 10 and 16). Solvent exposure was generally assessed with occupational history and/or air sampling. Similar to metals, chronic exposure to solvents in occupational settings was assessed with chronic sleep health outcomes such as OSA and insomnia. Of the 7 studies that evaluated solvent exposure and SDB or OSA, the majority relied on PSG or oximetry for outcome measurement and supported a positive association[96–100]. One study repeatedly measured a subset of solvent-exposed workers with OSA, reporting overall decreases in AHI following exposure cessation and increased AHI shortly after re-exposure within participants[96]. Studies of sleep quality were mixed but suggestive of an inverse association; a study by Lundberg et al reported a worse sleep quality and greater fragmentation but no difference in other insomnia-related symptoms or OSA as exposure increased[101]. Most findings of solvent exposure and sleep duration were null, but a case study reported increased sleep requirement among 26% of 19 adult males following occupational exposure to benzene and toluene[102]. Most self-reported data was collected using tools such as the NSC-60 and the Euroquest questionnaire. Overall, the quality of epidemiologic evidence for solvents was low (Supplemental File 10F), but the more rigorous studies suggested associations with sleep architecture, duration, and SDB among more highly exposed; the quality of evidence from case studies was fair (Supplemental Files 13 and 16).

DISCUSSION

This review systematically identified and evaluated 204 studies of exposure to chemical pollutants in relation to sleep health and disorders. Overall, particulate matter and nitrogen dioxide were linked with poor sleep quality, and tobacco smoke exposure in pediatric populations associated with SDB; dioxins, PCBs, and PBDEs were tied to sleep disruption and insomnia, and exposure to lead, mercury, and pesticides were also associated with insomnia. Solvents such as toluene were linked to disrupted sleep timing and SDB. Studies of conflict-related exposures relevant to GWI supported associations with poor sleep quality, insomnia, SDB, and altered sleep architecture. Pesticide exposure, particularly in pediatric populations, was not well explored. In general, the evidence presented in this review indicates environmental pollutants may be detrimental to sleep health and disorders among adult and pediatric populations.

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Synthesis and Potential Mechanisms

Numerous mechanistic pathways, some shared by multiple exposures, may link environmental chemical pollutant exposure to impaired sleep health and sleep disorders. Broadly, common characteristics of exposures associated with sleep outcomes include: acting on the cholinergic system, inducing oxidative stress or inflammation, altering neurotransmission, and endocrine disruption; other possible mechanisms that weren't explored in the literature include acting on components of the circadian clock or relevant pathways and hormones such as adenosine and melatonin.

Air pollution—Air pollution comprises a heterogenous mix of particulates and gases and may have acute or chronic effects on sleep health. PM and NO_2 were most consistently associated with poor sleep quality, and, in pediatric populations, SHS with insomnia, worse sleep quality, and SDB. PM may be composed of many different types of compounds and is categorized by size, with $PM_{2.5}$ the benchmark at which particles are small enough to pass directly from the lungs into the capillary blood supply and across the blood-brain barrier[103]. PM may disrupt sleep via systemic and/or pulmonary inflammation[104], but the evidence from included studies was mixed. In contrast to PM, gaseous components of air pollution may influence sleep by affecting neurotransmission. Experimental evidence suggests that O_3 exposure increases slow wave sleep (SWS)[105,106] and alters serotonin levels[107]. While 3 of 4 included studies with O_3 measures did not report associations, sleep was ascertained using questionnaires, ICD-10 codes, or text-mining, whereas the study that did report an association measured sleep patterns with actigraphy.

Ambient air pollution levels can either be averaged over a timescale of a year or longer to model the effects of chronic exposure or modeled on a day-to-day basis to evaluate acute effects on health. Chronic air pollution may be expected to influence development of sleep disorders, such as SDB and OSA, while acute air pollution may influence day-to-day changes in sleep patterns, such as insomnia symptoms (if insomnia not already present) and sleep duration. Included studies generally framed the exposure-outcome dynamic as long-term air pollutant exposure affecting chronic sleep conditions, although a few examined short-term dynamics[21,26].

Pulmonary impairment due to air pollution may contribute to SDB[108]. Experimental and epidemiological evidence links exposure to air pollution and inflammation[109,110], an established risk factor for sleep disturbances[111] and OSA[112]. For example, NO_2 has been linked both to pediatric asthma[113] and SDB[24], and studies previously reviewed in Liu et al[8] also supported a positive relationship between NO_2 and greater odds of OSA[114]. Furthermore, nicotine has well-established effects on sleep disruption, impairing initiation, maintenance, and duration[115], and affecting sleep architecture[115]. However, the impact of SHS on sleep outcomes is less conclusive[10,116]. Future studies should investigate co-exposures of air pollution and allergens as possible environmental drivers of pediatric OSA.

Conflict-related Exposures (Gulf War, WTC, Other)—During the Gulf War, military personnel may have been exposed to OP compounds (sarin) in addition to pyridostigmine

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bromide (a carbamate compound), pesticides, mustard gas, air pollution from oil well fires, and uranium. Exposures may have been acute or chronic during the period of deployment, with chronic impacts on sleep, measured by retrospective self-report. Numerous reports have concluded that psychological stress and/or post-traumatic stress disorder are unlikely causes of GWI[117], suggesting an environmental etiology. Many of the symptoms of GWI overlap with those of OP and nerve agent poisoning, and findings from a rat model of Gulf War exposures (permethrin, DEET, and pyridostigmine bromide over 4 weeks) support a causal relationship with GWI symptoms[118]. Both GWI symptoms[40] and exposure to sulfur mustard[50] were associated with higher self-reported OSA symptomology, suggesting a possible shared etiology with exposure to nerve agents and SDB. While not included in the review, a small study (n=18) of Gulf War veterans also reported greater SDB symptoms among those with GWI[119]. Likewise, sarin and cyclosarin are OP nerve agents that act on AChE, and experimental evidence supports a link between sarin and altered EEG patterns during sleep[120]. However, there remain knowledge gaps between other Gulf War-related exposures, such as with jet propulsion fuel 8 (JP-8)[121]; JP-8 has not been evaluated in relation to sleep outcomes but contains solvents, such as benzene, that have.

Similar to the heterogeneity of exposures in the Gulf War, first responders to the 2001 WTC attack may have had acute or chronic exposure to many different compounds during the rescue and clean-up operations, with exposure ascertained by duration and proximity to dust cloud and site. Four of the included WTC studies reported associations with OSA. The toxic dust cloud and smoke contained fine particulate matter, and, at lower concentrations, asbestos, heavy metals, dioxins, pesticides, phthalates, PBDEs, and polyaromatic hydrocarbons[122]. Chronic inflammation of the airways caused by this exposure could have increased the risk for OSA development, but the highly observed nature of these cohorts may have also increased case detection.

EDCs/Other—EDCs such as BPA, phthalates, PCBs, PBDEs, and dioxins, can bind to nuclear hormone receptors[123], such as estrogen receptors and the aryl hydrocarbon receptor (AhR). Sleep traits show sex differences and chronic disruption of these hormonal processes by EDCs could potentially elicit changes in sleep behavior[124,125]. For example, phthalates, used as plasticizers, are ubiquitous and have been shown to exert estrogenic activity[126]. Due to their short half-lives, phthalate metabolites measured in urine reflect acute exposure[127]; however, phthalates are common in consumer products, making chronic, repeated exposure likely. There were two cross-sectional analyses: one in NHANES where phthalate exposure measured at a single time point was associated with shorter sleep duration[58], and another where phthalate exposure (4 samples per individual pooled across time, 1 taken each week) was not associated with sleep disturbances, insomnia, or restless sleep in premenopausal and perimenopausal participants[124]. In contrast, dioxins, PCBs[128], and PBDEs were linked to insomnia and disrupted sleep. PBDEs are structurally similar to thyroid hormones[129] and could potentially influence sleep by disrupting thyroid hormone signaling[130]. Additionally, AhR activation may affect AChE and cholinergic signaling[131] as well as circadian clock function[132]. Two studies reported positive associations between BPA levels and short sleep duration[51] and OSA[53], and additional studies reported positive associations between PCBs, PBDEs, or dioxins and sleep

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impairment. As lipophilic pollutants, these compounds build up in body fat over time[133] and their accumulation, or body burden, may differ by age and body fat; however, adiposity may also increase OSA risk and is associated with short sleep duration[134], potentially confounding associations between lipophilic pollutants and sleep outcomes.

Metals—Metals may impact sleep by promoting inflammation, influencing neurotransmission, and/or affecting the cholinergic system. Experimental animal studies support a causal link between methylmercury and altered sleep-wake rhythms and sleep architecture[135,136], possibly through inhibition of muscarinic acetylcholine receptors[137]. Similarly, lead may affect sleep by influencing neurotransmission[138]; animal and in vitro studies have shown that lead exposure alters GABA release[139], and a study in monkeys reported that treatment with lead during infancy caused insomnia[140]. There were multiple studies of copper and/or selenium (in non-occupational settings), with inconclusive findings, and a few studies evaluated exposure to manganese, both a micronutrient and an occupational hazard for welders and metal workers. Manganese exposure has also been linked to sleep disturbances in human and animal studies[141]. High manganese levels in the brain may affect dopaminergic[142] and cholinergic neurotransmission, such as through AChE inhibition[143], and chronic manganese exposure in a rodent model causes sleep disruption[144]. However, epidemiologic studies of occupational manganese exposure with sleep duration and insomnia were null or inconclusive; one study of airborne manganese pollution reported shorter sleep duration and greater use of hypnotic medication among residents in areas with higher exposure[145]. Chronic exposure to metals, such as in occupational settings, would be expected to lead to long-lasting effects on sleep health, whereas acute exposure, such as in an accidental mercury poisoning, may cause either acute or chronic effects on sleep.

The overall evidence between metals and sleep outcomes were mixed, but results supported a link between lead or mercury exposure and insomnia; findings were more consistent in highly exposed groups and/or case studies of overt heavy metal poisoning.

Pesticides—Included studies evaluated exposure to organophosphate (OP), pyrethroid[146], and/or carbamate pesticides. OP compounds and carbamates inhibit AChE, an enzyme that breaks down acetylcholine to prevent further neurotransmission. As the cholinergic system is an important regulator of sleep, with acetylcholine playing a role in wakefulness and REM sleep[147], acute or chronic inhibition of AChE by pesticides may affect sleep through cholinergic signaling[148]. Altered cholinergic signaling, implicated in upper airway collapsibility and OSA[149], may also be a pathway by which environmental exposures, such as pesticides, influence SDB. Experimental studies in humans support a relationship between exposure to AChE agents and insomnia[150], and animal studies also support a causal relationship between pesticide exposure and sleep impairment. For example, rats exposed to chlorpyrifos, a commonly used OP pesticide, during early development had higher sleep apnea index and lower diaphragm acetylcholinesterase (AChE) activity later in life, compared to unexposed rats[151]. Chlorpyrifos-exposed rats also exhibited decreased sleep and sleep spindles[152]. Interestingly, AChE inhibitor drugs

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are also used in treating Alzheimer's disease symptoms; among the documented side effects of these drugs are insomnia[153] and altered sleep architecture[154].

In addition to inhibiting AChE, carbamates may affect sleep by binding to melatonin receptors, phase-shifting circadian rhythms, and altering pineal melatonin synthesis[155–157]; an *in silico* binding study demonstrated the ability of carbaryl and carbofuran to bind the MT1 and MT2 receptors[156]. Similar to metals, the overall epidemiologic evidence for pesticides were mixed, whereas case study findings tended to support an association; however, case studies may be more likely to report the presence, rather than absence, of sleeping problems. Additionally, case studies tended to evaluate acute, high exposure or outright poisoning, whereas epidemiologic studies relied on farm work as proxy or self-reported data on prior pesticide poisoning symptoms, for example. Despite possible mechanistic pathways, the overall epidemiologic evidence for pesticides and sleep disruption was weak due to lack of device-based exposure and/or outcome measurements.

Solvents—Solvents, such as benzene, toluene, and xylenes, are well-established occupational hazards; chronic or acute exposure can lead to solvent-induced encephalopathy. Solvents are mucosal irritants which can cross the blood-brain barrier and act as central depressants, promoting GABAergic and glycinergic signaling[158]. Benzene, one among the included solvents, has been associated with sleep disturbances in both human and animal studies[159]. Other solvents not among the included literature, such as trichloroethylene, have also been linked to sleepiness in humans and sleep disturbances in animal studies[160]. Animal studies have also shown that toluene exposure alters monoamines and SWS in rats[161–163]. Solvents may influence OSA development by promoting inflammation and increasing upper airway collapsibility[158]. However, the association between occupational solvent exposure and OSA is unclear within the realm of current workplace standards, and large prospective cohort studies of solvent exposure should consider including validated sleep and SDB measures as health outcomes.

Limitations of this Review

The findings presented in this review are not without limitations. This review focused on peer-reviewed literature, which may not capture all relevant public health information (such as from public health agencies). Due to the heterogeneity of pollutants and how sleep outcomes were measured, we did not perform a meta-analysis of the results or compare effect estimates. In some cases, multiple exposures or sleep outcomes were grouped together as a single item, limiting our ability to tease apart specific health associations. Additionally, while we present results on SHS/ETS and conflict-related exposures, the search strategy was not designed with these exposures in mind; in particular, the findings presented on SHS/ETS are not a comprehensive presentation of the literature, and prior systematic reviews on this topic should be sought for further coverage on the topic[9,10].

Current Gaps and Future Directions

The majority of reviewed studies relied on objective measures of exposure and one or limited measures of sleep heath using questionnaire data. While self-reported sleep outcomes may capture elements of sleep that are important for health, such measures

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are also prone to measurement error and reporting bias[164]. Questionnaires such as the PSQI or STOP-BANG have been evaluated for use in studies of sleep, but a significant portion of included studies did not employ validated instruments and/or relied upon a single sleep-related question as an outcome. Additionally, insomnia is a self-reported condition that is diagnosed clinically by interview and patient response, but most studies of questionnaire-assessed insomnia did not consider chronicity or associated daytime impairment – features needed to diagnose insomnia as a clinical disorder[5]. Among device-based measures, PSG is considered the gold standard for sleep measurement, but it can be expensive, time-consuming, and affected by the "first night effect". Home-based PSG systems or PSG-like systems, and/or actigraphy, which measures movement, may be less onerous and more cost-effective alternatives. However, device specifications should be thoroughly considered in study design[165].

However, there are also limitations of subjective or objective exposure measurement. The measurement accuracy of self-reported exposure via a single questionnaire item or calculated with a job exposure matrix may differ by chemical type, setting, route of exposure, and duration (acute or chronic). Likewise, even objectively measured exposures may be limited by choice of sampling location (e.g., air measured in a room versus using a body-worn device), biomatrix (e.g., urine, blood, hair, teeth, etc.), timing, duration (acute or chronic), and exposure compound (e.g., parent compound or metabolite). Future studies should clearly discuss the hypothesized pollutant-sleep relationship and explain the reasoning behind study design decisions in exposure and outcome ascertainment.

Other gaps include the extent to which sleeping and circadian phase influence pollutant exposure and vulnerability, bidirectionality, and co-exposures. Vulnerability to toxicants may differ by time of the day[166] and there may be rhythmicity in biomarkers, but unfortunately most studies did not provide timing of sample collection or measurement or take circadian phase into account. Surprisingly, only one study included a measure of melatonin[50], and few studies investigated sleep timing. Likewise, co-exposures were rarely considered in the included literature; for example, areas with air pollution due to vehicle emissions may consequently also have higher burden of light at night and noise pollution. Furthermore, the role of the indoor sleep environment on pollutant exposure was underexplored. In summary, future research would benefit from integrating measures of circadian phase in the exposure/outcome measurement or as an interaction in the exposure-outcome relationship and timing of measurement and sample collection in study design. Shift work or other exposures which may disrupt circadian rhythms should also be considered.

Environmental pollutant exposures tend to be patterned by socioeconomic status, race, and/or ethnicity. Due to systematic racism and redlining policies that resulted in segregated, under-resourced neighborhoods, marginalized communities in the U.S. are more likely to be exposed to environmental chemical pollutants and adverse characteristics of the built environment[14]. Individuals living in these communities are more likely to have exposure to air pollution[12] and adverse sleep health is more prevalent among marginalized groups[167], yet few studies examine environmental pollutants as contributors to sleep disparities. Future studies in the U.S. and globally should aim to expand study samples

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to enroll sufficient sample sizes of historically marginalized or excluded populations, in addition to including measures of environmental pollutant exposure in sleep health disparities research studies.

This review of current evidence of environmental exposures and sleep outcomes in adults and children reveals the need for future research to consider a life course approach. Children may be especially vulnerable to pollutants' effects on brain development and sleep health; harmful environmental exposures potentially differ by age, and pediatric populations are of special interest because infants and children have no or underdeveloped detoxification systems. One such gap in knowledge is the lack of studies investigating pediatric pesticide exposure and sleep outcomes. Exposures during childhood can influence health trajectory; applying a life course approach with particular focus on environmental exposures during vulnerable developmental windows and critical periods[168] may be informative for research in environmental determinants of sleep health. Thus, identifying the environmental factors that lead to disrupted sleep in both childhood and adulthood can inform public health efforts to improve sleep health and address disparities in sleep and related morbidity[169,170].

Conclusions

Results document associations between air pollutants, conflict-related exposures, EDCs, metals, pesticides, and solvents with sleep health. Possible biological pathways underlying pollutant-sleeprelationships included cholinergic signaling, inflammation, neurotransmission, and hormonal signaling. To improve collaboration and scientific discovery, researchers can contribute to the sleep research community by sharing well-annotated data in repositories such as the National Sleep Research Resource (NSRR, https://sleepdata.org/). There is a large gap in current knowledge around environmental pollutant exposure and pediatric sleep health, especially regarding pesticides. Overall, future studies should be robustly designed to evaluate environmental exposures and sleep health, with device-based measures of exposures and outcomes. Longitudinal study design incorporating measures of actigraphy and wearable devices, including representation of diverse backgrounds, as well as clear reporting and data availability will advance our understanding of the environmental contributions to sleep health across the life course.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

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Data Availability:

The search terms, detailed tables with study information, and extracted data used for plotting the RoB figures are provided in Supplemental Files.

References

[1]. Borbély AA, Daan S, Wirz-Justice A, Deboer T. The two-process model of sleep regulation: a reappraisal. J Sleep Res 2016;25:131–43. 10.1111/jsr.12371. [PubMed: 26762182]

[2]. Buysse DJ. Sleep health: can we define it? Does it matter? Sleep 2014;37:9–17. 10.5665/sleep.3298. [PubMed: 24470692]

[3]. Wheaton AG, Claussen AH. Short Sleep Duration Among Infants, Children, and Adolescents Aged 4 Months-17 Years - United States, 2016–2018. MMWR Morb Mortal Wkly Rep 2021;70:1315–21. 10.15585/mmwr.mm7038a1. [PubMed: 34555000]

[4]. Lyons MM, Bhatt NY, Pack AI, Magalang UJ. Global burden of sleep-disordered breathing and its implications. Respirology 2020;25:690–702. 10.1111/resp.13838. [PubMed: 32436658]

[5]. Roth T. Insomnia: definition, prevalence, etiology, and consequences. J Clin Sleep Med 2007;3:S7–10. 10.5664/jcsm.26929. [PubMed: 17824495]

[6]. Zielinski MR, McKenna JT, McCarley RW. Functions and mechanisms of sleep. AIMS Neurosci 2016;3:67–104. 10.3934/Neuroscience.2016.1.67. [PubMed: 28413828]

[7]. Johnson DA, Billings ME, Hale L. Environmental determinants of insufficient sleep and sleep disorders: implications for population health. Curr Epidemiol Rep 2018;5:61–9. 10.1007/s40471-018-0139-y. [PubMed: 29984131]

[8]. Liu J, Wu T, Liu Q, Wu S, Chen J-C. Air pollution exposure and adverse sleep health across the life course: A systematic review. Environ Pollut 2020;262:114263. 10.1016/j.envpol.2020.114263. [PubMed: 32443219]

[9]. Safa F, Chaiton M, Mahmud I, Ahmed S, Chu A. The association between exposure to second-hand smoke and sleep disturbances: A systematic review and meta-analysis. Sleep Health 2020;6:702–14. 10.1016/j.sleh.2020.03.008. [PubMed: 32446663]

[10]. Jara SM, Benke JR, Lin SY, Ishman SL. The association between secondhand smoke and sleep-disordered breathing in children: a systematic review. Laryngoscope 2015;125:241–7. 10.1002/lary.24833. [PubMed: 25130300]

[11]. Schwartz DA, Vinnikov D, Blanc PD. Occupation and Obstructive Sleep Apnea: A Meta-Analysis. J Occup Environ Med 2017;59:502–8. 10.1097/JOM.0000000000001008. [PubMed: 28598928]

[12]. Liu J, Clark LP, Bechle MJ, Hajat A, Kim S-Y, Robinson AL, et al. Disparities in Air Pollution Exposure in the United States by Race/Ethnicity and Income, 1990–2010. Environ Health Perspect 2021;129:127005. 10.1289/EHP8584. [PubMed: 34908495]

[13]. Jackson CL, Redline S, Emmons KM. Sleep as a potential fundamental contributor to disparities in cardiovascular health. Annu Rev Public Health 2015;36:417–40. 10.1146/annurevpublhealth-031914-122838. [PubMed: 25785893]

[14]. Billings ME, Cohen RT, Baldwin CM, Johnson DA, Palen BN, Parthasarathy S, et al. Disparities in sleep health and potential intervention models: A focused review. Chest 2021;159:1232–40. 10.1016/j.chest.2020.09.249. [PubMed: 33007324]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[15]. Murad MH, Sultan S, Haffar S, Bazerbachi F. Methodological quality and synthesis of case series and case reports. BMJ Evid Based Med 2018;23:60–3. 10.1136/bmjebm-2017-110853.

[16]. Accinelli RA, Llanos O, López LM, Matayoshi S, Oros YP, Kheirandish-Gozal L, et al. Caregiver perception of sleep-disordered breathing-associated symptoms in children of rural Andean communities above 4000 masl with chronic exposure to biomass fuel. Sleep Med 2015;16:723–8. 10.1016/j.sleep.2015.02.536. [PubMed: 26002760]

[17]. Wang Y, Liu X, Chen G, Tu R, Abdulai T, Qiao D, et al. Association of long-term exposure to ambient air pollutants with prolonged sleep latency: The Henan Rural Cohort Study. Environ Res 2020;191:110116. 10.1016/j.envres.2020.110116. [PubMed: 32846171]

[18]. He Y, Liu X, Luo Z, Wang Y, Fan K, Tu R, et al. Long-term exposure to ambient PM1 strengthened the association of depression/anxiety symptoms with poor sleep quality: The Henan Rural Cohort study. Ecotoxicol Environ Saf 2021;211:111932. 10.1016/j.ecoenv.2021.111932. [PubMed: 33476852]

[19]. Chen G, Xiang H, Mao Z, Huo W, Guo Y, Wang C, et al. Is long-term exposure to air pollution associated with poor sleep quality in rural China? Environ Int 2019;133:105205. 10.1016/j.envint.2019.105205. [PubMed: 31639600]

[20]. Yu Z, Wei F, Wu M, Lin H, Shui L, Jin M, et al. Association of long-term exposure to ambient air pollution with the incidence of sleep disorders: A cohort study in China. Ecotoxicol Environ Saf 2021;211:111956. 10.1016/j.ecoenv.2021.111956. [PubMed: 33493724]

[21]. Li W, Bertisch SM, Mostofsky E, Vgontzas A, Mittleman MA. Associations of daily weather and ambient air pollution with objectively assessed sleep duration and fragmentation: a prospective cohort study. Sleep Med 2020;75:181–7. 10.1016/j.sleep.2020.06.029. [PubMed: 32858358]

[22]. Pelgrims I, Devleesschauwer B, Guyot M, Keune H, Nawrot TS, Remmen R, et al. Association between urban environment and mental health in Brussels, Belgium. BMC Public Health 2021;21:635. 10.1186/s12889-021-10557-7. [PubMed: 33794817]

[23]. Li L, Zhang W, Xie L, Jia S, Feng T, Yu H, et al. Effects of atmospheric particulate matter pollution on sleep disorders and sleep duration: a cross-sectional study in the UK biobank. Sleep Med 2020;74:152–64. 10.1016/j.sleep.2020.07.032. [PubMed: 32861010]

[24]. Manzoni F, Tentoni S, Nosetti L, Niespolo A, Monestier A, Montomoli C. INTERNATIONAL PERSPECTIVES. Journal of Environmental Health 2018;81:E1.

[25]. Gokdemir Y, Civelek E, Cakir B, Demir A, Kocabas CN, Ikizoglu NB, et al. Prevalence of sleep-disordered breathing and associated risk factors in primary school children in urban and rural environments. Sleep Breath 2021;25:915–22. 10.1007/s11325-020-02206-x. [PubMed: 33030645]

[26]. Heyes A, Zhu M. Air pollution as a cause of sleeplessness: Social media evidence from a panel of Chinese cities. Journal of Environmental Economics and Management 2019;98:102247. 10.1016/j.jeem.2019.07.002.

[27]. Tang M, Li D, Liew Z, Wei F, Wang J, Jin M, et al. The association of short-term effects of air pollution and sleep disorders among elderly residents in China. Sci Total Environ 2020;708:134846. 10.1016/j.scitotenv.2019.134846. [PubMed: 31780155]

[28]. Chen C, Liu GG, Sun Y, Gu D, Zhang H, Yang H, et al. Association between household fuel use and sleep quality in the oldest-old: Evidence from a propensity-score matched case-control study in Hainan, China. Environ Res 2020;191:110229. 10.1016/j.envres.2020.110229. [PubMed: 32950513]

[29]. Ekici M, Ekici A, Keles H, Akin A, Karlidag A, Tunckol M, et al. Risk factors and correlates of snoring and observed apnea. Sleep Med 2008;9:290–6. 10.1016/j.sleep.2007.04.018. [PubMed: 17658298]

[30]. Gan D, Huang D, Yang J, Zhang L, Ou S, Feng Y, et al. Assessment of kitchen emissions using a backpropagation neural network model based on urinary hydroxy polycyclic aromatic hydrocarbons. Environ Pollut 2020;265:114915. 10.1016/j.envpol.2020.114915. [PubMed: 32535415]

[31]. Djeddi D, Stephan-Blanchard E, Léké A, Ammari M, Delanaud S, Lemaire-Hurtel A-S, et al. Effects of Smoking Exposure in Infants on Gastroesophageal Reflux as a Function of the

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

Sleep-Wakefulness State. J Pediatr 2018;201:147–53. 10.1016/j.jpeds.2018.05.057. [PubMed: 30041936]

[32]. Veronda AC, Irish LA, Delahanty DL. Effect of smoke exposure on young adults' sleep quality. Nurs Health Sci 2020;22:57–63. 10.1111/nhs.12644. [PubMed: 31424168]

[33]. Sánchez T, Rojas C, Casals M, Bennett JT, Gálvez C, Betancur C, et al. [Prevalence and risk factors for sleep-disordered breathing in chilean schoolchildren]. Rev Chil Pediatr 2018;89:718–25. 10.4067/S0370-41062018005000902. [PubMed: 30725060]

[34]. Groner JA, Nicholson L, Huang H, Bauer JA. Secondhand Smoke Exposure and Sleep-Related Breathing Problems in Toddlers. Acad Pediatr 2019;19:835–41. 10.1016/j.acap.2019.03.008. [PubMed: 30959225]

[35]. Yolton K, Xu Y, Khoury J, Succop P, Lanphear B, Beebe DW, et al. Associations between secondhand smoke exposure and sleep patterns in children. Pediatrics 2010;125:e261–8. 10.1542/peds.2009-0690. [PubMed: 20083521]

[36]. Davis CL, Tingen MS, Jia J, Sherman F, Williams CF, Bhavsar K, et al. Passive smoke exposure and its effects on cognition, sleep, and health outcomes in overweight and obese children. Child Obes 2016;12:119–25. 10.1089/chi.2015.0083. [PubMed: 26812049]

[37]. Fukuda K, Nisenbaum R, Stewart G, Thompson WW, Robin L, Washko RM, et al. Chronic multisymptom illness affecting Air Force veterans of the Gulf War. JAMA 1998;280:981–8. 10.1001/jama.280.11.981. [PubMed: 9749480]

[38]. Kroenke K, Koslowe P, Roy M. Symptoms in 18,495 Persian Gulf War veterans. Latency of onset and lack of association with self-reported exposures. J Occup Environ Med 1998;40:520–8. 10.1097/00043764-199806000-00004. [PubMed: 9636932]

[39]. Steele L. Prevalence and patterns of Gulf War illness in Kansas veterans: association of symptoms with characteristics of person, place, and time of military service. Am J Epidemiol 2000;152:992–1002. 10.1093/aje/152.10.992. [PubMed: 11092441]

[40]. Chao LL, Abadjian LR, Esparza IL, Reeb R. Insomnia severity, subjective sleep quality, and risk for obstructive sleep apnea in veterans with gulf war illness. Mil Med 2016;181:1127–34. 10.7205/MILMED-D-15-00474. [PubMed: 27612364]

[41]. Sunderram J, Weintraub M, Black K, Alimokhtari S, Twumasi A, Sanders H, et al. Chronic rhinosinusitis is an independent risk factor for OSA in world trade center responders. Chest 2019;155:375–83. 10.1016/j.chest.2018.10.015. [PubMed: 30739642]

[42]. de la Hoz RE, Aurora RN, Landsbergis P, Bienenfeld LA, Afilaka AA, Herbert R. Snoring and obstructive sleep apnea among former World Trade Center rescue workers and volunteers. J Occup Environ Med 2010;52:29–32. 10.1097/JOM.0b013e3181c2bb18. [PubMed: 20042888]

[43]. de la Hoz RE, Mallea JM, Kramer SJ, Bienenfeld LA, Wisnivesky JP, Aurora RN. Polysomnographic diagnoses among former world trade center rescue workers and volunteers. Arch Environ Occup Health 2012;67:239–42. 10.1080/19338244.2012.725230. [PubMed: 23074981]

[44]. Ahuja S, Zhu Z, Shao Y, Berger KI, Reibman J, Ahmed O. Obstructive sleep apnea in community members exposed to world trade center dust and fumes. J Clin Sleep Med 2018;14:735–43. 10.5664/jcsm.7094. [PubMed: 29735001]

[45]. Sunderram J, Udasin I, Kelly-McNeil K, Ko S, Cepeda C, Marroccoli B, et al. Unique features of obstructive sleep apnea in World Trade Center responders with aerodigestive disorders. J Occup Environ Med 2011;53:975–80. 10.1097/JOM.0b013e3182305282. [PubMed: 21866046]

[46]. Webber MP, Lee R, Soo J, Gustave J, Hall CB, Kelly K, et al. Prevalence and incidence of high risk for obstructive sleep apnea in World Trade Center-exposed rescue/recovery workers. Sleep Breath 2011;15:283–94. 10.1007/s11325-010-0379-7. [PubMed: 20593281]

[47]. Glaser MS, Shah N, Webber MP, Zeig-Owens R, Jaber N, Appel DW, et al. Obstructive sleep apnea and World Trade Center exposure. J Occup Environ Med 2014;56 Suppl 10:S30–4. 10.1097/JOM.0000000000000283. [PubMed: 25285973]

[48]. Duffy FH, Burchfiel JL, Bartels PH, Gaon M, Sim VM. Long-term effects of an organophosphate upon the human electroencephalogram. Toxicol Appl Pharmacol 1979;47:161–76. 10.1016/0041-008X(79)90083-8. [PubMed: 425115]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[49]. Nakajima T, Ohta S, Fukushima Y, Yanagisawa N. Sequelae of sarin toxicity at one and three years after exposure in Matsumoto, Japan. J Epidemiol 1999;9:337–43. 10.2188/jea.9.337. [PubMed: 10616267]

[50]. Mousavi SS, Vahedi E, Shohrati M, Panahi Y, Parvin S. Nocturnal serum melatonin levels in sulfur mustard exposed patients with sleep disorders. J R Army Med Corps 2017;163:411–5. 10.1136/jramc-2016-000677. [PubMed: 29146723]

[51]. Beydoun HA, Beydoun MA, Jeng HA, Zonderman AB, Eid SM. Bisphenol-A and Sleep Adequacy among Adults in the National Health and Nutrition Examination Surveys. Sleep 2016;39:467–76. 10.5665/sleep.5466. [PubMed: 26446109]

[52]. Ye X, Wong L-Y, Bishop AM, Calafat AM. Variability of urinary concentrations of bisphenol A in spot samples, first morning voids, and 24-hour collections. Environ Health Perspect 2011;119:983–8. 10.1289/ehp.1002701. [PubMed: 21406337]

[53]. Erden ES, Genc S, Motor S, Ustun I, Ulutas KT, Bilgic HK, et al. Investigation of serum bisphenol A, vitamin D, and parathyroid hormone levels in patients with obstructive sleep apnea syndrome. Endocrine 2014;45:311–8. 10.1007/s12020-013-0022-z. [PubMed: 23904340]

[54]. Ji H, Liang H, Wang Z, Miao M, Wang X, Zhang X, et al. Associations of prenatal exposures to low levels of Polybrominated Diphenyl Ether (PBDE) with thyroid hormones in cord plasma and neurobehavioral development in children at 2 and 4 years. Environ Int 2019;131:105010. 10.1016/j.envint.2019.105010. [PubMed: 31326823]

[55]. Neuberger M. Chloracne and morbidity after dioxin exposure (preliminary results). Toxicol Lett 1998;96–97:347–50. 10.1016/S0378-4274(98)00092-7.

[56]. Fitzgerald EF, Weinstein AL, Youngblood LG, Standfast SJ, Melius JM. Health effects three years after potential exposure to the toxic contaminants of an electrical transformer fire. Arch Environ Health 1989;44:214–21. 10.1080/00039896.1989.9935886. [PubMed: 2506840]

[57]. Kondo H, Tanio K, Nagaura Y, Nagayoshi M, Mitoma C, Furue M, et al. Sleep disorders among Yusho patients highly intoxicated with dioxin-related compounds: A 140-case series. Environ Res 2018;166:261–8. 10.1016/j.envres.2018.05.033. [PubMed: 29908457]

[58]. Sears CG, Braun JM. Urinary phthalate metabolite concentrations and adolescent sleep duration. Environ Epidemiol 2021;5:e134. 10.1097/EE9.0000000000000134. [PubMed: 33870010]

[59]. Sińczuk-Walczak H, Szymczak M, Raźniewska G, Matczak W, Szymczak W. Effects of occupational exposure to aluminum on nervous system: clinical and electroencephalographic findings. Int J Occup Med Environ Health 2003;16:301–10. [PubMed: 14964639]

[60]. Handra CM, Ghita I, Ulmeanu A, Enache A-M, Epureanu F, Coman OA, et al. Depressive Clinical Manifestations Associated with Professional Aluminum Exposure. Rev Chim 2019;70:2162–7. 10.37358/RC.19.6.7297.

[61]. Smith RG, Vorwald AJ, Patil LS, Mooney TF. Effects of exposure to mercury in the manufacture of chlorine. Am Ind Hyg Assoc J 1970;31:687–700. 10.1080/0002889708506315. [PubMed: 5275968]

[62]. Decharat S Urinary Mercury Levels Among Workers in E-waste Shops in Nakhon Si Thammarat Province, Thailand. J Prev Med Public Health 2018;51:196–204. 10.3961/jpmph.18.049. [PubMed: 30071707]

[63]. Lönnroth E-C, Shahnavaz H Amalgam in dentistry A health hazard for dental personnel? Int J Occup Saf Ergon 1997;3:151–60. 10.1080/10803548.1997.11076372. [PubMed: 10602603]

[64]. Naimi-Akbar A, Svedberg P, Alexanderson K, Carlstedt-Duke B, Ekstrand J, Englund GS. Health-related quality of life and symptoms in patients with experiences of health problems related to dental restorative materials. Community Dent Oral Epidemiol 2013;41:163–72. 10.1111/cdoe.12002. [PubMed: 22963525]

[65]. Hilt B, Svendsen K, Syversen T, Aas O, Qvenild T. Occurrence of cognitive and neurological symptoms in norwegian dentists. Saf Health Work 2011;2:176–82. 10.5491/SHAW.2011.2.2.176. [PubMed: 22953200]

[66]. Nilsson B, Gerhardsson L, Nordberg GF. Urine mercury levels and associated symptoms in dental personnel. Sci Total Environ 1990;94:179–85. 10.1016/0048-9697(90)90169-u. [PubMed: 2363035]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[67]. Siblerud RL, Motl J, Kienholz E. Psychometric evidence that mercury from silver dental fillings may be an etiological factor in depression, excessive anger, and anxiety. Psychol Rep 1994;74:67–80. 10.2466/pr0.1994.74.1.67. [PubMed: 8153237]

[68]. Awad el Karim MA, Hamed AS, Elhaimi YA, Osman Y. Effects of exposure to lead among lead-acid battery factory workers in Sudan. Arch Environ Health 1986;41:261–5. 10.1080/00039896.1986.9938342. [PubMed: 3767435]

[69]. Malekirad AA, Oryan S, Fani A, Babapor V, Hashemi M, Baeeri M, et al. Study on clinical and biochemical toxicity biomarkers in a zinc-lead mine workers. Toxicol Ind Health 2010;26:331–7. 10.1177/0748233710365697. [PubMed: 20371635]

[70]. Liu J, Liu X, Pak V, Wang Y, Yan C, Pinto-Martin J, et al. Early blood lead levels and sleep disturbance in preadolescence. Sleep 2015;38:1869–74. 10.5665/sleep.5230. [PubMed: 26194570]

[71]. Kruger DJ, Kodjebacheva GD, Cupal S. Poor tap water quality experiences and poor sleep quality during the Flint, Michigan Municipal Water Crisis. Sleep Health 2017;3:241–3. 10.1016/j.sleh.2017.05.007. [PubMed: 28709509]

[72]. Jansen EC, Dunietz GL, Dababneh A, Peterson KE, Chervin RD, Baek J, et al. Cumulative childhood lead levels in relation to sleep during adolescence. J Clin Sleep Med 2019;15:1443–9. 10.5664/jcsm.7972. [PubMed: 31596209]

[73]. Kordas K, Casavantes KM, Mendoza C, Lopez P, Ronquillo D, Rosado JL, et al. The association between lead and micronutrient status, and children's sleep, classroom behavior, and activity. Arch Environ Occup Health 2007;62:105–12. 10.3200/AEOH.62.2.105-112. [PubMed: 18316268]

[74]. Brockhaus A, Dolgner R, Ewers U, Krämer U, Soddemann H, Wiegand H. Intake and health effects of thallium among a population living in the vicinity of a cement plant emitting thallium containing dust. Int Arch Occup Environ Health 1981;48:375–89. 10.1007/BF00378686. [PubMed: 6913543]

[75]. Aprea MC, Nuvolone D, Petri D, Voller F, Bertelloni S, Aragona I. Human biomonitoring to assess exposure to thallium following the contamination of drinking water. PLoS ONE 2020;15:e0241223. 10.1371/journal.pone.0241223. [PubMed: 33119651]

[76]. Ghaderi A, NasehGhafoori P, Rasouli-Azad M, Sehat M, Mehrzad F, Nekuei M, et al. Examining of thallium in cigarette smokers. Biol Trace Elem Res 2018;182:224–30. 10.1007/s12011-017-1107-y. [PubMed: 28766107]

[77]. Li Y, Jing D, Xiao Y, Huang X, Shen M. Patient-Reported Outcomes of Arsenic-Related Skin Lesions in China. Biomed Res Int 2020;2020:6195975. 10.1155/2020/6195975. [PubMed: 33015173]

[78]. Sińczuk-Walczak H, Szymczak M, Hałatek T. Effects of occupational exposure to arsenic on the nervous system: clinical and neurophysiological studies. Int J Occup Med Environ Health 2010;23:347–55. 10.2478/v10001-010-0034-3. [PubMed: 21306980]

[79]. Malin AJ, Bose S, Busgang SA, Gennings C, Thorpy M, Wright RO, et al. Fluoride exposure and sleep patterns among older adolescents in the United States: a cross-sectional study of NHANES 2015–2016. Environ Health 2019;18:106. 10.1186/s12940-019-0546-7. [PubMed: 31818308]

[80]. Gump BB, Gabrikova E, Bendinskas K, Dumas AK, Palmer CD, Parsons PJ, et al. Low-level mercury in children: associations with sleep duration and cytokines TNF-α and IL-6. Environ Res 2014;134:228–32. 10.1016/j.envres.2014.07.026. [PubMed: 25173056]

[81]. Jansen EC, Hector EC, Goodrich JM, Cantoral A, Téllez Rojo MM, Basu N, et al. Mercury exposure in relation to sleep duration, timing, and fragmentation among adolescents in Mexico City. Environ Res 2020;191:110216. 10.1016/j.envres.2020.110216. [PubMed: 32956656]

[82]. Bose-O'Reilly S, Schierl R, Nowak D, Siebert U, William JF, Owi FT, et al. A preliminary study on health effects in villagers exposed to mercury in a small-scale artisanal gold mining area in Indonesia. Environ Res 2016;149:274–81. 10.1016/j.envres.2016.04.007. [PubMed: 27174690]

[83]. Centers for Disease Control and Prevention (CDC). Lead poisoning in bridge demolition workers-Georgia, 1992. MMWR Morb Mortal Wkly Rep 1993;42:388–90. [PubMed: 8492795]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[84]. Rae CE, Bell CN, Elliott CE, Shannon M. Ten cases of acute lead intoxication among bridge workers in Louisiana. DICP 1991;25:932–7. 10.1177/106002809102500905. [PubMed: 1949969]

[85]. Bowler RM, Roels HA, Nakagawa S, Drezgic M, Diamond E, Park R, et al. Dose-effect relationships between manganese exposure and neurological, neuropsychological and pulmonary function in confined space bridge welders. Occup Environ Med 2007;64:167–77. 10.1136/oem.2006.028761. [PubMed: 17018581]

[86]. Josephs KA, Ahlskog JE, Klos KJ, Kumar N, Fealey RD, Trenerry MR, et al. Neurologic manifestations in welders with pallidal MRI T1 hyperintensity. Neurology 2005;64:2033–9. 10.1212/01.WNL.0000167411.93483.A1. [PubMed: 15888601]

[87]. Metcalf DR, Holmes JH. VII. Toxicology and physiology. EEG, psychological, and neurological alterations in humans with organophosphorus exposure. Ann N Y Acad Sci 1969;160:357–65. 10.1111/j.1749-6632.1969.tb15857.x. [PubMed: 5257403]

[88]. Baumert BO, Carnes MU, Hoppin JA, Jackson CL, Sandler DP, Freeman LB, et al. Sleep apnea and pesticide exposure in a study of US farmers. Sleep Health 2018;4:20–6. 10.1016/j.sleh.2017.08.006. [PubMed: 29332674]

[89]. Li J, Hao Y, Tian D, He S, Sun X, Yang H. Relationship between cumulative exposure to pesticides and sleep disorders among greenhouse vegetable farmers. BMC Public Health 2019;19:373. 10.1186/s12889-019-6712-6. [PubMed: 30943937]

[90]. Rubin C, Esteban E, Kieszak S, Hill RH, Dunlop B, Yacovac R, et al. Assessment of human exposure and human health effects after indoor application of methyl parathion in Lorain County, Ohio, 1995–1996. Environ Health Perspect 2002;110 Suppl 6:1047–51. 10.1289/ehp.02110s61047.

[91]. Branch RA, Jacqz E. Subacute neurotoxicity following long-term exposure to carbaryl. Am J Med 1986;80:741–5. 10.1016/0002-9343(86)90837-5. [PubMed: 3083676]

[92]. Bakhsh K, Ahmad N, Kamran MA, Hassan S, Abbas Q, Saeed R, et al. Occupational hazards and health cost of women cotton pickers in Pakistani Punjab. BMC Public Health 2016;16:961. 10.1186/s12889-016-3635-3. [PubMed: 27618928]

[93]. Beshwari MMM, Bener A, Ameen A, Al-Mehdi AM, Ouda HZ, Pasha MAH. Pesticide-related health problems and diseases among farmers in the United Arab Emirates. International Journal of Environmental Health Research 1999;9:213–21.

[94]. El-Desouky MA, Hassan RZ. Exposure to pesticides in Kuwait. J Kuwait Med Assoc 1978;12:215–27.

[95]. Malekirad AA, Faghih M, Mirabdollahi M, Kiani M, Fathi A, Abdollahi M. Neurocognitive, mental health, and glucose disorders in farmers exposed to organophosphorus pesticides. Archives of Industrial Hygiene and Toxicology 2013;64:1–8. 10.2478/10004-1254-64-2013-2296. [PubMed: 23705196]

[96]. Monstad P, Mellgren SI, Sulg IA. The clinical significance of sleep apnoea in workers exposed to organic solvents: implications for the diagnosis of organic solvent encephalopathy. J Neurol 1992;239:195–8. 10.1007/BF00839139. [PubMed: 1597686]

[97]. Godderis L, Dours G, Laire G, Viaene MK. Sleep apnoeas and neurobehavioral effects in solvent exposed workers. Int J Hyg Environ Health 2011;214:66–70. 10.1016/j.ijheh.2010.08.006. [PubMed: 20843741]

[98]. Laire G, Viaene MK, Veulemans H, Masschelein R, Nemery B. Nocturnal oxygen desaturation, as assessed by home oximetry, in long-term solvent-exposed workers. Am J Ind Med 1997;32:656–64. 10.1002/(sici)1097-0274(199712)32:6<656::aid-ajim12>3.0.co;2-0. [PubMed: 9358923]

[99]. Ulfberg J, Carter N, Talbäck M, Edling C. Occupational exposure to organic solvents and sleep-disordered breathing. Neuroepidemiology 1997;16:317–26. 10.1159/000109704. [PubMed: 9430132]

[100]. Sekkal S, Casas L, Haddam N, Bouhacina L, Scheers H, Taleb A, et al. Sleep disturbances and neurotoxicity in workers exposed to hydrocarbons. An observational study from Algeria. Am J Ind Med 2016;59:129–36. 10.1002/ajim.22561. [PubMed: 26725468]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[101]. Lundberg I, Michélsen H, Nise G, Hogstedt C, Högberg M, Alfredsson L, et al. Neuropsychiatric function of housepainters with previous long-term heavy exposure to organic solvents. Scand J Work Environ Health 1995;21 Suppl 1:1–44. [PubMed: 8824938]

[102]. Kraut A, Lilis R, Marcus M, Valciukas JA, Wolff MS, Landrigan PJ. Neurotoxic effects of solvent exposure on sewage treatment workers. Arch Environ Health 1988;43:263–8. 10.1080/00039896.1988.10545947. [PubMed: 3415351]

[103]. Elder A, Gelein R, Silva V, Feikert T, Opanashuk L, Carter J, et al. Translocation of inhaled ultrafine manganese oxide particles to the central nervous system. Environ Health Perspect 2006;114:1172–8. 10.1289/ehp.9030. [PubMed: 16882521]

[104]. Jiang X, Han Y, Qiu X, Chai Q, Zhang H, Chen X, et al. Organic Components of Personal PM2.5 Exposure Associated with Inflammation: Evidence from an Untargeted Exposomic Approach. Environ Sci Technol 2021;55:10589–96. 10.1021/acs.est.1c02023. [PubMed: 34297563]

[105]. Alfaro-Rodríguez A, González-Piña R. Ozone-induced paradoxical sleep decrease is related to diminished acetylcholine levels in the medial preoptic area in rats. Chem Biol Interact 2005;151:151–8. 10.1016/j.cbi.2004.10.001. [PubMed: 15733536]

[106]. Paz C, Bazan-Perkins B. Sleep-wake disorganization in cats exposed to ozone. Neurosci Lett 1992;140:270–2. 10.1016/0304-3940(92)90118-q. [PubMed: 1501789]

[107]. González-Piña R, Alfaro-Rodríguez A. Ozone exposure alters 5-hydroxy-indole-acetic acid contents in dialysates from dorsal raphe and medial preoptic area in freely moving rats. Relationships with simultaneous sleep disturbances. Chem Biol Interact 2003;146:147–56. 10.1016/s00092797(03)00103-0. [PubMed: 14597128]

[108]. Zanobetti A, Redline S, Schwartz J, Rosen D, Patel S, O'Connor GT, et al. Associations of PM10 with sleep and sleep-disordered breathing in adults from seven U.S. urban areas. Am J Respir Crit Care Med 2010;182:819–25. 10.1164/rccm.200912-1797OC. [PubMed: 20508218]

[109]. Chuang K-J, Chan C-C, Su T-C, Lee C-T, Tang C-S. The effect of urban air pollution on inflammation, oxidative stress, coagulation, and autonomic dysfunction in young adults. Am J Respir Crit Care Med 2007;176:370–6. 10.1164/rccm.200611-1627OC. [PubMed: 17463411]

[110]. Sun Q, Wang A, Jin X, Natanzon A, Duquaine D, Brook RD, et al. Long-term air pollution exposure and acceleration of atherosclerosis and vascular inflammation in an animal model. JAMA 2005;294:3003–10. 10.1001/jama.294.23.3003. [PubMed: 16414948]

[111]. Irwin MR, Olmstead R, Carroll JE. Sleep Disturbance, Sleep Duration, and Inflammation: A Systematic Review and Meta-Analysis of Cohort Studies and Experimental Sleep Deprivation. Biol Psychiatry 2016;80:40–52. 10.1016/j.biopsych.2015.05.014. [PubMed: 26140821]

[112]. Huang T, Goodman M, Li X, Sands SA, Li J, Stampfer MJ, et al. C-reactive Protein and Risk of OSA in Four US Cohorts. Chest 2021;159:2439–48. 10.1016/j.chest.2021.01.060. [PubMed: 33529772]

[113]. Anenberg SC, Mohegh A, Goldberg DL, Kerr GH, Brauer M, Burkart K, et al. Long-term trends in urban NO2 concentrations and associated paediatric asthma incidence: estimates from global datasets. Lancet Planet Health 2022;6:e49–58. 10.1016/S2542-5196(21)00255-2. [PubMed: 34998460]

[114]. Billings ME, Gold D, Szpiro A, Aaron CP, Jorgensen N, Gassett A, et al. The Association of Ambient Air Pollution with Sleep Apnea: The Multi-Ethnic Study of Atherosclerosis. Ann Am Thorac Soc 2019;16:363–70. 10.1513/AnnalsATS.201804-248OC. [PubMed: 30571166]

[115]. Jaehne A, Loessl B, Bárkai Z, Riemann D, Hornyak M. Effects of nicotine on sleep during consumption, withdrawal and replacement therapy. Sleep Med Rev 2009;13:363–77. 10.1016/j.smrv.2008.12.003. [PubMed: 19345124]

[116]. Davila EP, Lee DJ, Fleming LE, LeBlanc WG, Arheart K, Dietz N, et al. Sleep disorders and secondhand smoke exposure in the U.S. population. Nicotine Tob Res 2010;12:294–9. 10.1093/ntr/ntp193. [PubMed: 20133380]

[117]. White RF, Steele L, O'Callaghan JP, Sullivan K, Binns JH, Golomb BA, et al. Recent research on Gulf War illness and other health problems in veterans of the 1991 Gulf War: Effects of toxicant exposures during deployment. Cortex 2016;74:449–75. 10.1016/j.cortex.2015.08.022. [PubMed: 26493934]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[118]. Parihar VK, Hattiangady B, Shuai B, Shetty AK. Mood and memory deficits in a model of Gulf War illness are linked with reduced neurogenesis, partial neuron loss, and mild inflammation in the hippocampus. Neuropsychopharmacology 2013;38:2348–62. 10.1038/npp.2013.158. [PubMed: 23807240]

[119]. Amin MM, Belisova Z, Hossain S, Gold MS, Broderick JE, Gold AR. Inspiratory airflow dynamics during sleep in veterans with Gulf War illness: a controlled study. Sleep Breath 2011;15:333–9. 10.1007/s11325-010-0386-8. [PubMed: 20703820]

[120]. Shih TM, Hulet SW, McDonough JH. The effects of repeated low-dose sarin exposure. Toxicol Appl Pharmacol 2006;215:119–34. 10.1016/j.taap.2006.02.003. [PubMed: 16556454]

[121]. Baldwin CM, Figueredo AJ, Wright LS, Wong SS, Witten ML. Repeated aerosol-vapor JP-8 jet fuel exposure affects neurobehavior and neurotransmitter levels in a rat model. J Toxicol Environ Health Part A 2007;70:1203–13. 10.1080/15287390701380872.

[122]. Lippmann M, Cohen MD, Chen L-C. Health effects of World Trade Center (WTC) Dust: An unprecedented disaster's inadequate risk management. Crit Rev Toxicol 2015;45:492–530. 10.3109/10408444.2015.1044601. [PubMed: 26058443]

[123]. Diamanti-Kandarakis E, Bourguignon J-P, Giudice LC, Hauser R, Prins GS, Soto AM, et al. Endocrine-disrupting chemicals: an Endocrine Society scientific statement. Endocr Rev 2009;30:293–342. 10.1210/er.2009-0002. [PubMed: 19502515]

[124]. Hatcher KM, Smith RL, Chiang C, Li Z, Flaws JA, Mahoney MM. Association of phthalate exposure and endogenous hormones with self-reported sleep disruptions: results from the Midlife Women's Health Study. Menopause 2020;27:1251–64. 10.1097/GME.0000000000001614. [PubMed: 33110041]

[125]. Mong JA, Cusmano DM. Sex differences in sleep: impact of biological sex and sex steroids. Philos Trans R Soc Lond B Biol Sci 2016;371:20150110. 10.1098/rstb.2015.0110. [PubMed: 26833831]

[126]. Harris CA, Henttu P, Parker MG, Sumpter JP. The estrogenic activity of phthalate esters in vitro. Environ Health Perspect 1997;105:802–11. 10.1289/ehp.97105802. [PubMed: 9347895]

[127]. Toxicological Profile for Di(2-Ethylhexyl)Phthalate (DEHP). Atlanta (GA): Agency for Toxic Substances and Disease Registry (US); 2022.

[128]. Agency for Toxic Substances and Disease Registry (US), editor. Toxicological Profile for Polychlorinated Biphenyls (PCBs) (Agency for toxic substances and disease registry (ATSDR) toxicological profiles). Atlanta, GA: 2000.

[129]. Darnerud PO, Eriksen GS, Jóhannesson T, Larsen PB, Viluksela M. Polybrominated diphenyl ethers: occurrence, dietary exposure, and toxicology. Environ Health Perspect 2001;109 Suppl 1:49–68. 10.1289/ehp.01109s149. [PubMed: 11250805]

[130]. Shekhar S, Hall JE, Klubo-Gwiezdzinska J. The hypothalamic pituitary thyroid axis and sleep. Current Opinion in Endocrine and Metabolic Research 2021;17:8–14. 10.1016/j.coemr.2020.10.002. [PubMed: 34322645]

[131]. Xie HQ, Xu H-M, Fu H-L, Hu Q, Tian W-J, Pei X-H, et al. AhR-mediated effects of dioxin on neuronal acetylcholinesterase expression in vitro. Environ Health Perspect 2013;121:613–8. 10.1289/ehp.1206066. [PubMed: 23426015]

[132]. Wang C, Zhang Z-M, Xu C-X, Tischkau SA. Interplay between Dioxin-mediated signaling and circadian clock: a possible determinant in metabolic homeostasis. Int J Mol Sci 2014;15:11700–12. 10.3390/ijms150711700. [PubMed: 24987953]

[133]. La Merrill M, Emond C, Kim MJ, Antignac J-P, Le Bizec B, Clément K, et al. Toxicological function of adipose tissue: focus on persistent organic pollutants. Environ Health Perspect 2013;121:162–9. 10.1289/ehp.1205485. [PubMed: 23221922]

[134]. Jehan S, Zizi F, Pandi-Perumal SR, Wall S, Auguste E, Myers AK, et al. Obstructive sleep apnea and obesity: implications for public health. Sleep Med Disord 2017;1.

[135]. Algarve TD, Assmann CE, Aigaki T, da Cruz IBM. Parental and preimaginal exposure to methylmercury disrupts locomotor activity and circadian rhythm of adult Drosophila melanogaster. Drug Chem Toxicol 2020;43:255–65. 10.1080/01480545.2018.1485689. [PubMed: 30033776]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[136]. Arito H, Hara N, Torii S. Effect of methylmercury chloride on sleep-waking rhythms in rats. Toxicology 1983;28:335–45. 10.1016/0300-483X(83)90007-0. [PubMed: 6648981]

[137]. Basu N, Kwan M, Chan HM. Mercury but not organochlorines inhibits muscarinic cholinergic receptor binding in the cerebrum of ringed seals (Phoca hispida). J Toxicol Environ Health Part A 2006;69:1133–43. 10.1080/15287390500362394.

[138]. Lidsky TI, Schneider JS. Lead neurotoxicity in children: basic mechanisms and clinical correlates. Brain 2003;126:5–19. 10.1093/brain/awg014. [PubMed: 12477693]

[139]. Abadin H, Ashizawa A, Stevens Y-W, Llados F, Diamond G, Sage G, et al. Toxicological profile for lead. Atlanta (GA): Agency for Toxic Substances and Disease Registry (US); 2007. 10.1201/9781420061888_ch106.

[140]. Allen JR, McWey PJ, Suomi SJ. Pathobiological and behavioral effects of lead intoxication in the infant rhesus monkey. Environ Health Perspect 1974;7:239–46. 10.1289/ehp.747239. [PubMed: 4208658]

[141]. Agency for Toxic Substances and Disease Registry (US), editor. Toxicological Profile for Manganese (Agency for toxic substances and disease registry (ATSDR) toxicological profiles). Atlanta, GA: 2012.

[142]. O'Neal SL, Zheng W. Manganese toxicity upon overexposure: a decade in review. Curr Environ Health Rep 2015;2:315–28. 10.1007/s40572-015-0056-x. [PubMed: 26231508]

[143]. Harischandra DS, Ghaisas S, Zenitsky G, Jin H, Kanthasamy A, Anantharam V, et al. Manganese-Induced Neurotoxicity: New Insights Into the Triad of Protein Misfolding, Mitochondrial Impairment, and Neuroinflammation. Front Neurosci 2019;13:654. 10.3389/fnins.2019.00654. [PubMed: 31293375]

[144]. Bouabid S, Fifel K, Benazzouz A, Lakhdar-Ghazal N. Consequences of manganese intoxication on the circadian rest-activity rhythms in the rat. Neuroscience 2016;331:13–23. 10.1016/j.neuroscience.2016.06.016. [PubMed: 27316552]

[145]. Bowler RM, Adams SW, Wright CW, Kim Y, Booty A, Colledge M, et al. Medication use associated with exposure to manganese in two Ohio towns. Int J Environ Health Res 2016;26:483–96. 10.1080/09603123.2016.1194381. [PubMed: 27295281]

[146]. Wolansky MJ, Harrill JA. Neurobehavioral toxicology of pyrethroid insecticides in adult animals: a critical review. Neurotoxicol Teratol 2008;30:55–78. 10.1016/j.ntt.2007.10.005. [PubMed: 18206347]

[147]. Brown RE, Basheer R, McKenna JT, Strecker RE, McCarley RW. Control of sleep and wakefulness. Physiol Rev 2012;92:1087–187. 10.1152/physrev.00032.2011. [PubMed: 22811426]

[148]. Stallones L, Beseler CL. Assessing the connection between organophosphate pesticide poisoning and mental health: A comparison of neuropsychological symptoms from clinical observations, animal models and epidemiological studies. Cortex 2016;74:405–16. 10.1016/j.cortex.2015.10.002. [PubMed: 26654853]

[149]. Zhu L, Chamberlin NL, Arrigoni E. Muscarinic Inhibition of Hypoglossal Motoneurons: Possible Implications for Upper Airway Muscle Hypotonia during REM Sleep. J Neurosci 2019;39:7910–9. 10.1523/JNEUROSCI.0461-19.2019. [PubMed: 31420456]

[150]. Grob D, Harvey JC. Effects in man of the anticholinesterase compound sarin (isopropyl methyl phosphonofluoridate). J Clin Invest 1958;37:350–68. 10.1172/JCI103615. [PubMed: 13513766]

[151]. Darwiche W, Gay-Quéheillard J, Delanaud S, El Khayat El Sabbouri H, Khachfe H, Joumaa W, et al. Impact of chronic exposure to the pesticide chlorpyrifos on respiratory parameters and sleep apnea in juvenile and adult rats. PLoS ONE 2018;13:e0191237. 10.1371/journal.pone.0191237. [PubMed: 29357379]

[152]. Timofeeva OA, Gordon CJ. Changes in EEG power spectra and behavioral states in rats exposed to the acetylcholinesterase inhibitor chlorpyrifos and muscarinic agonist oxotremorine. Brain Res 2001;893:165–77. 10.1016/s0006-8993(00)03309-6. [PubMed: 11223004]

[153]. Rogers SL, Doody RS, Mohs RC, Friedhoff LT. Donepezil improves cognition and global function in Alzheimer disease: a 15-week, double-blind, placebo-controlled study. Donepezil Study Group. Arch Intern Med 1998;158:1021–31. 10.1001/archinte.158.9.1021. [PubMed: 9588436]

Author Manuscript Author Manuscript Author Manuscript Author Manuscript

[154]. Moraes W dos S, Poyares DR, Guilleminault C, Ramos LR, Bertolucci PHF, Tufik S. The effect of donepezil on sleep and REM sleep EEG in patients with Alzheimer disease: a double-blind placebo-controlled study. Sleep 2006;29:199–205. 10.1093/sleep/29.2.199. [PubMed: 16494088]

[155]. Glatfelter GC, Jones AJ, Rajnarayanan RV, Dubocovich ML. Pharmacological actions of carbamate insecticides at mammalian melatonin receptors. J Pharmacol Exp Ther 2021;376:306–21. 10.1124/jpet.120.000065. [PubMed: 33203660]

[156]. Popovska-Gorevski M, Dubocovich ML, Rajnarayanan RV. Carbamate insecticides target human melatonin receptors. Chem Res Toxicol 2017;30:574–82. 10.1021/acs.chemrestox.6b00301. [PubMed: 28027439]

[157]. Attia A, Reiter R, Nonaka K, Mostafa M, Soliman S, Elsebae A. Carbaryl-induced changes in indoleamine synthesis in the pineal gland and its effects on nighttime serum melatonin concentrations. Toxicology 1991;65:305–14. 10.1016/0300-483X(91)90089-J. [PubMed: 1704155]

[158]. Bushnell PJ, Shafer TJ, Bale AS, Boyes WK, Simmons JE, Eklund C, et al. Developing an exposuredose-response model for the acute neurotoxicity of organic solvents: overview and progress on in vitro models and dosimetry. Environ Toxicol Pharmacol 2005;19:607–14. 10.1016/j.etap.2004.12.026. [PubMed: 21783533]

[159]. Agency for Toxic Substances and Disease Registry (US), editor. Toxicological Profile for Benzene (Agency for toxic substances and disease registry (ATSDR) toxicological profiles). Atlanta, GA: 2007.

[160]. Agency for Toxic Substances and Disease Registry (US), editor. Toxicological Profile forTrichloroethylene. Atlanta, GA: 2019.

[161]. Alfaro-Rodríguez A, Bueno-Nava A, González-Piña R, Arch-Tirado E, Vargas-Sánchez J, Avila-Luna A. Chronic exposure to toluene changes the sleep-wake pattern and brain monoamine content in rats. Acta Neurobiol Exp (Wars) 2011;71:183–92. [PubMed: 21731073]

[162]. Arito H, Tsuruta H, Oguri M. Changes in sleep and wakefulness following single and repeated exposures to toluene vapor in rats. Arch Toxicol 1988;62:76–80. 10.1007/BF00316262. [PubMed: 3190461]

[163]. Hisanaga N, Takeuchi Y. Changes in sleep cycle and EEG of rats exposed to 4000 ppm toluene for four weeks. Ind Health 1983;21:153–64. 10.2486/indhealth.21.153. [PubMed: 6629857]

[164]. Jackson CL, Patel SR, Jackson WB, Lutsey PL, Redline S. Agreement between self-reported and objectively measured sleep duration among white, black, Hispanic, and Chinese adults in the United States: Multi-Ethnic Study of Atherosclerosis. Sleep 2018;41. 10.1093/sleep/zsy057.

[165]. Mazzotti DR, Haendel MA, McMurry JA, Smith CJ, Buysse DJ, Roenneberg T, et al. Sleep and circadian informatics data harmonization: a workshop report from the Sleep Research Society and Sleep Research Network. Sleep 2022;45. 10.1093/sleep/zsac002.

[166]. Dallmann R, Okyar A, Lévi F. Dosing-Time Makes the Poison: Circadian Regulation and Pharmacotherapy. Trends Mol Med 2016;22:430–45. 10.1016/j.molmed.2016.03.004. [PubMed: 27066876]

[167]. Johnson DA, Jackson CL, Williams NJ, Alcántara C. Are sleep patterns influenced by race/ethnicity - a marker of relative advantage or disadvantage? Evidence to date. Nat Sci Sleep 2019;11:79–95. 10.2147/NSS.S169312. [PubMed: 31440109]

[168]. Landrigan PJ, Kimmel CA, Correa A, Eskenazi B. Children's health and the environment: public health issues and challenges for risk assessment. Environ Health Perspect 2004;112:257–65. 10.1289/ehp.6115. [PubMed: 14754581]

[169]. Gaston SA, Jackson CL. Strengthening the case for early-life interventions to address racial/ethnic sleep disparities across the life-course using an exposome approach. Sleep 2021;44. 10.1093/sleep/zsab182.

[170]. Johnson DA, Ohanele C, Alcántara C, Jackson CL. The need for social and environmental determinants of health research to understand and intervene on racial/ethnic disparities in obstructive sleep apnea. Clin Chest Med 2022;43:199–216. 10.1016/j.ccm.2022.02.002. [PubMed: 35659019]

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Practice Points

1. Most studies were conducted in adult populations and evaluated heavy metals, such as mercury or lead, or air pollution
2. Possible underlying mechanisms between chemical pollutant exposure and sleep relate to cholinergic signaling, inflammation, neurotransmission, and hormonal signaling;
3. Evidence was mixed, but suggested associations between air pollution, exposures related to the Gulf War, dioxin and dioxin-like compounds, and heavy metals with sleep impairment.

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Research Agenda

Future research should focus on:

1. Inclusion of pediatric and historically marginalized or excluded populations in environmental sleep research;
2. Objective measurement of both environmental exposures and sleep, as well as mechanistic investigation;
3. Bidirectionality of sleeping and circadian phase with pollutant exposure and vulnerability;
4. The impacts of heavy metals, air pollution, and pesticides on sleep, as well as co-exposures.

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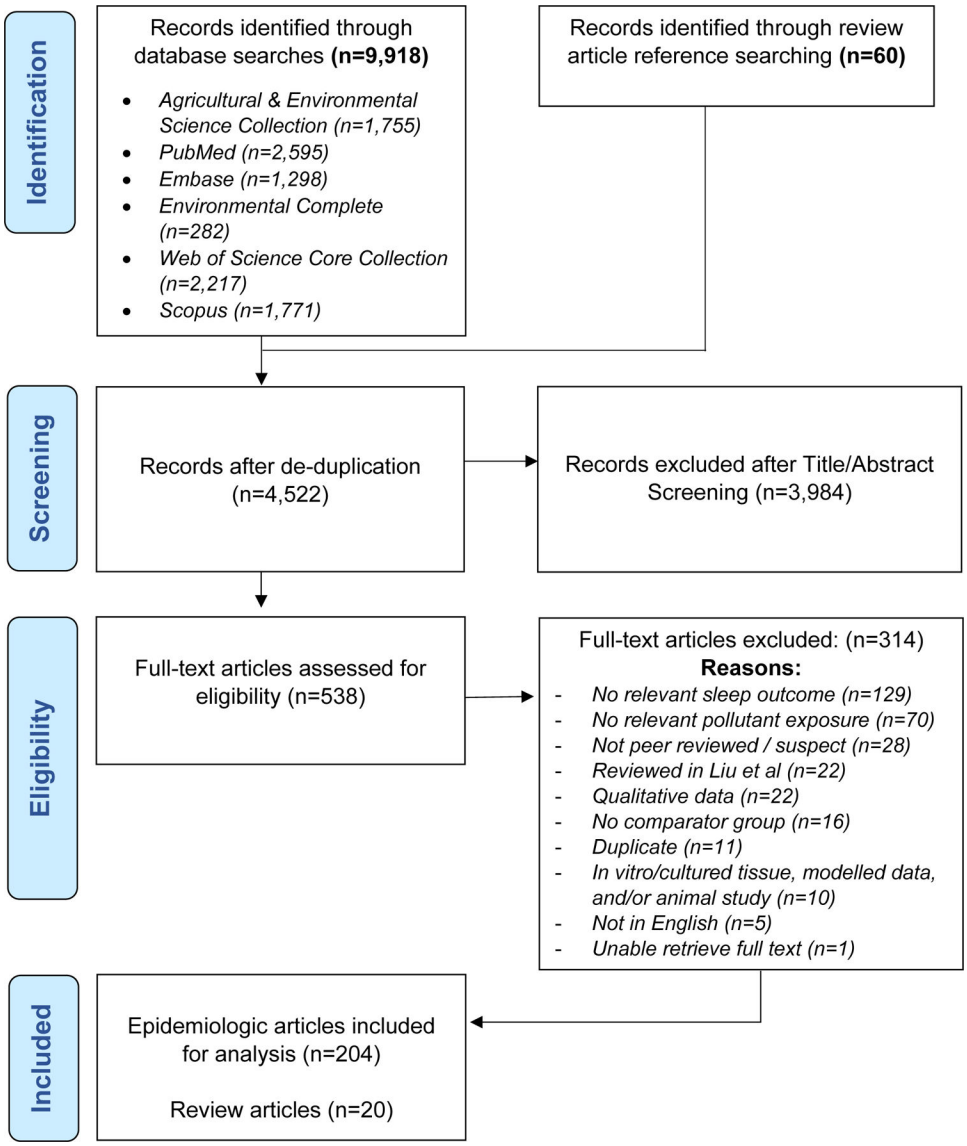

Figure 1.

Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) flow chart detailing screening and inclusion of studies in this review.